Diabetes — Scene 3 Incretins: GLP-1 / GIP Harbor

1. GLP-1 ship — the agents

WHAT YOU SEE

Purple-sailed ship 'GLP-1' with banner 'LIRA-SEMA-DULA-EXENA.'

WHAT IT ENCODES

GLP-1 receptor agonists (the '-tides'): liraglutide and semaglutide (daily/weekly), dulaglutide (weekly), exenatide. They augment glucose-dependent insulin secretion, suppress glucagon, slow gastric emptying, and promote central satiety, giving robust A1c lowering (~1-1.5%) with substantial weight loss. Because insulin release is glucose-dependent, monotherapy carries low intrinsic hypoglycemia risk. Mnemonic: the '-tides' rise in the harbor (peptides, injected; oral semaglutide is the exception).

BOARD TRAP

Wrong answer: 'start a sulfonylurea for this obese T2DM patient with ASCVD.' Discriminator — sulfonylureas cause weight gain and hypoglycemia and have no CV benefit; a GLP-1 RA is preferred when obesity and atherosclerotic CV disease coexist.

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2. GIP/GLP-1 dual ship — tirzepatide

WHAT YOU SEE

Second ship 'GLP-1 / GIP' with banner 'TIRZEPATIDE.'

WHAT IT ENCODES

Tirzepatide is a dual GIP + GLP-1 receptor agonist (weekly injection) delivering the greatest A1c reduction (~2%) and weight loss (up to ~15-20%) of any non-insulin agent. It shares the class GI side-effect profile (nausea, vomiting) and the medullary thyroid carcinoma / MEN2 boxed warning. Titrate slowly from 2.5 mg weekly to limit GI intolerance.

BOARD TRAP

Wrong answer: 'tirzepatide is a pure GLP-1 agonist.' Discriminator — it is a dual incretin (GIP + GLP-1); this dual mechanism is why it outperforms single-agonist GLP-1 RAs on weight and glycemic endpoints.

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3. Pramlintide rowboat

WHAT YOU SEE

Small rowboat banner 'PRAMLINTIDE.'

WHAT IT ENCODES

Pramlintide is an amylin analog (not an incretin) given subcutaneously before meals as an adjunct to mealtime insulin in T1DM or T2DM. It slows gastric emptying, suppresses glucagon, and promotes satiety. Key safety point: when added to insulin it markedly raises hypoglycemia risk, so reduce prandial insulin by ~50% when starting; never mix in the same syringe with insulin.

BOARD TRAP

Wrong answer: 'pramlintide is an incretin mimetic like the GLP-1 agonists.' Discriminator — it is an amylin analog co-secreted with insulin from beta cells; it is used WITH insulin (mealtime), not as a standalone incretin therapy.

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4. Weight loss crate

WHAT YOU SEE

Crane lowering crate 'WEIGHT LOSS'; happy slim figure.

WHAT IT ENCODES

Substantial weight loss is the defining class benefit of incretin therapies: semaglutide ~10-15% and tirzepatide up to ~15-20% of body weight (high-dose semaglutide and tirzepatide are also FDA-approved for obesity). This contrasts sharply with weight-gaining agents (sulfonylureas, TZDs, insulin). On boards, 'obese T2DM needing weight loss' is the trigger phrase for a GLP-1 RA.

BOARD TRAP

Wrong answer: 'add basal insulin to drive down the A1c in this obese patient.' Discriminator — insulin causes weight gain and hypoglycemia; in an obese T2DM patient without insulinopenia, a weight-losing GLP-1 RA/tirzepatide is favored before insulin intensification.

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5. Slow gastric emptying

WHAT YOU SEE

Stomach + clock building: 'slow gastric.'

WHAT IT ENCODES

GLP-1 RAs slow gastric emptying, blunting the postprandial glucose spike and producing early satiety; this same effect drives the dose-limiting nausea, vomiting, and early-satiety side effects (usually transient with slow titration). Clinically relevant: hold GLP-1 RAs before elective procedures/anesthesia because retained gastric contents raise aspiration risk.

BOARD TRAP

Wrong answer: 'continue the GLP-1 agonist in a patient with diabetic gastroparesis.' Discriminator — slowed gastric emptying worsens gastroparesis; gastroparesis is a contraindication, and persistent severe nausea should prompt drug discontinuation.

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6. Pancreatitis risk

WHAT YOU SEE

Pancreas figure lying down on dock — pancreatitis warning.

WHAT IT ENCODES

Incretin therapies (GLP-1 RAs, DPP-4 inhibitors, tirzepatide) carry a pancreatitis warning. New severe persistent abdominal pain radiating to the back warrants stopping the drug and checking lipase; do not restart if pancreatitis is confirmed. Avoid initiating in patients with a prior history of pancreatitis.

BOARD TRAP

Wrong answer: 'this severe epigastric pain on a GLP-1 agonist is just expected GI nausea — keep the drug.' Discriminator — persistent severe abdominal pain (vs the mild self-limited nausea of slowed emptying) should raise suspicion for drug-associated pancreatitis; stop the agent and evaluate.

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7. MEN2 / MTC contraindication

WHAT YOU SEE

Thyroid building 'MTC' / 'MEN2' with red X.

WHAT IT ENCODES

Boxed warning: GLP-1 RAs and tirzepatide are contraindicated with a personal or family history of medullary thyroid carcinoma (MTC) or MEN2 syndrome, based on rodent C-cell tumors. Always screen this history before prescribing. DPP-4 inhibitors do NOT carry this contraindication.

BOARD TRAP

Wrong answer: 'prescribe semaglutide for this obese diabetic whose mother had medullary thyroid cancer.' Discriminator — family history of MTC/MEN2 is an absolute contraindication to GLP-1 RAs and tirzepatide; choose a non-incretin agent (e.g., SGLT2 inhibitor) instead.

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8. MEN2 crate (cross-ref)

WHAT YOU SEE

Crate labeled 'MEN2 / MTC' with red X near dock.

WHAT IT ENCODES

Reinforces the MTC / MEN2 contraindication shared by all GLP-1 RAs and the dual GIP/GLP-1 agonist tirzepatide. Remember the wider MEN2 picture (MEN2A: MTC + pheochromocytoma + parathyroid hyperplasia; MEN2B: MTC + pheo + mucosal neuromas/marfanoid habitus) — any of these flags the incretin contraindication.

BOARD TRAP

Wrong answer: 'a DPP-4 inhibitor is contraindicated in MEN2 just like GLP-1 agonists.' Discriminator — the MTC/MEN2 boxed warning applies to GLP-1 RAs and tirzepatide, NOT to DPP-4 inhibitors; gliptins remain an option in these patients.

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9. Best for obese + ASCVD

WHAT YOU SEE

Barrel/sign 'OBESE + ASCVD.'

WHAT IT ENCODES

The board-preferred niche for GLP-1 RAs: T2DM with obesity AND established atherosclerotic cardiovascular disease (or high ASCVD risk). ADA guidance favors a GLP-1 RA with proven CV benefit (e.g., semaglutide, dulaglutide, liraglutide) in this group, independent of A1c or metformin use.

BOARD TRAP

Wrong answer: 'in T2DM with ASCVD, glycemic target alone dictates therapy.' Discriminator — agent selection is driven by comorbidity (ASCVD → GLP-1 RA with CV benefit; HF/CKD → SGLT2 inhibitor), not solely by the A1c number.

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10. CV benefit fountain

WHAT YOU SEE

Fountain 'CV' with feather/quill.

WHAT IT ENCODES

GLP-1 RAs with proven CV benefit (liraglutide, semaglutide, dulaglutide) reduce major adverse cardiovascular events (MACE: CV death, MI, stroke) in high-risk T2DM. Tirzepatide also shows cardiometabolic benefit. This atherosclerotic/MACE benefit complements the SGLT2-inhibitor strength in heart-failure and renal outcomes.

BOARD TRAP

Wrong answer: 'GLP-1 agonists are first-line for reducing heart failure hospitalizations.' Discriminator — GLP-1 RAs reduce atherosclerotic MACE; the agent class with dominant heart-failure-hospitalization and CKD-progression benefit is the SGLT2 inhibitors.
